

Advanced Dynare

Chapter X *Forecasting*

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Outline

1. Introduction
2. Stochastic models without parameter uncertainty
 - Unconditional forecasts
 - Conditional forecasts
 - Calibrated smoother
3. Stochastic models without parameter uncertainty
4. Perfect foresight conditional forecasts

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Motivation

- Structural models are often used for forecasting exercises
- Smets and Wouters (2007): performance on par with time series techniques
- Dynare provides a variety of tools to perform such exercises
- Important conceptual distinctions:
 1. conditional vs. unconditional forecasting
 2. stochastic vs. perfect foresight forecasting
 3. point forecast vs. density forecast

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Forecasting based on recursive policy rules

- Perturbation solutions with `stoch_simul` yield recursive policy rules for the endogenous variables x_{t-1} of the form

$$y_t = g(x_{t-1}, \varepsilon_t | \theta), \quad (1)$$

where

- x_{t-1} are endogenous state variables
 - ε_t are exogenous state variables
 - θ are structural models parameters
-
- Given initial conditions x_T and sequence of future shocks $\varepsilon_{T+i}, i = \{1, \dots, h\}$, we can compute future path of all endogenous variables y_t
→ conditional forecasts require selecting specific ε_{T+i}
 - Drawing from distribution of shocks for ε_{T+i} allows generating density forecasts that take shock uncertainty into account
→ no parameter uncertainty as calibration θ is given

Basic unconditional forecast: Dynare code I

rbc_basic_forecast.mod

- Consider again the basic RBC model with AR(1) technology shocks:

```
model;
  c + k = exp(A)*k(-1)^alpha + (1-delta)*k(-1);
  c^(-gamma) =
    beta*c(+1)^(-gamma)*(alpha*exp(A(+1))*k^(alpha-1) + 1-delta);
  A= rho*A(-1)+epsilon;
end;

steady_state_model;
  A = 0;
  k = ((1-beta*(1-delta))/(beta*alpha*exp(A)))^(1/(alpha-1));
  c = exp(A)*k^alpha-delta*k;
end;
steady;
```

Basic unconditional forecast: Dynare code II

rbc_basic_forecast.mod

- forecast command currently only allow first order approximation
→ should change for Dynare 7
- Define the shock distribution and solve the model at first order:

```
shocks;  
    var epsilon=0.01^2;  
end;  
  
stoch_simul(order=1,irf=0);
```


Basic unconditional forecast: Dynare code III

rbc_basic_forecast.mod

- We can set the initial condition for the state variable k to be 5% below its steady state value using `histval`-block
- The `forecast` command allows computing a 100-period density forecast with 95% confidence intervals

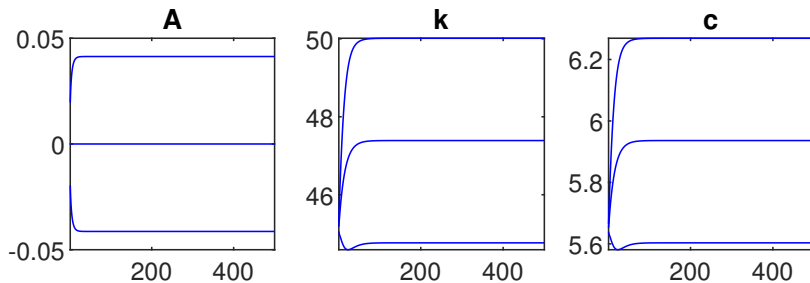
```
ik = varlist_indices('k',M_.endo_names);
kstar = oo_.steady_state(ik);

histval;
    k(0) = kstar*0.95;
end;

forecast(periods=100,conf_sig=0.95);
```

Basic unconditional forecast: Dynare output

rbc_basic_forecast.mod



- Results are stored in the `oo_.forecast` field

Deterministic exogenous shocks

rbc_basic_forecast_varexo_det.mod

- Perturbation solutions up to order=2 allow having deterministic exogenous shocks in addition to stochastic shocks (implementation details at [Dynare Wiki](#))
→ deterministic exogenous shocks become known with certainty at time T
- varexo_det statement declares deterministic exogenous shocks

```
varexo_det eps_det;  
...  
model;  
...  
A= rho*A(-1)+epsilon + eps_det;  
end;
```

Deterministic exogenous shocks: Dynare code

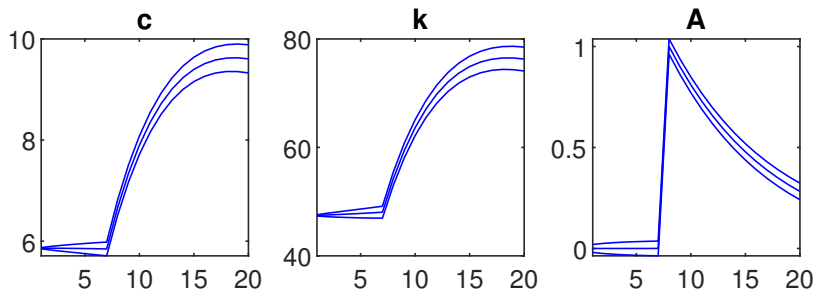
rbc_basic_forecast_varexo_det.mod

- perfect foresight-type syntax allows specifying the realization period and value of the deterministic shocks

```
shocks;  
  var epsilon=0.01^2;  
  var eps_det; periods 8; values 1;  
end;  
  
stoch_simul(order=1,irf=0);  
  
forecast(periods=20,conf_sig=0.95);
```

Deterministic exogenous shocks: Dynare output

rbc_basic_forecast_varexo_det.mod



- Deterministic shock occurs in period 8 and is known with certainty at the time of forecasting
- Works like a news shock (e.g. Beaudry and Portier 2006)
→ forward-looking agents react to it already before it materializes

Conditional forecasts

- Policy-makers are often interested in conditional forecasts, i.e. forecasts given future paths of some endogenous variable(s)
- Example: what happens if we set nominal interest rate to zero for the next 2 quarters?
- Achieving given future path of endogenous variables requires solving for shocks that generate this path
→ requires as many shocks as controlled endogenous variables
- Having more shocks and computing most likely linear combination can be achieved with Kalman filter-type algorithm (e.g. Bańbura et al. 2015), but is not (yet) implemented in Dynare

Conditional forecasts: setup

- Relies on state-space form from first-order perturbation solution:

$$y_t = T y_{t-1} + R \varepsilon_t \quad (2)$$

- Without loss of generality, at time t both y_t and ε_t can be split into controlled (top) and uncontrolled (bottom):

$$\begin{pmatrix} y_{1,t} \\ y_{2,t} \end{pmatrix} = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix}, \quad (3)$$

where $y_{1,t}$ are “controlled” endogenous variables and $\varepsilon_{1,t}$ are controlled shocks

- Goal: solve linear system for $\varepsilon_{1,t}$ to get desired $y_{1,t}$
→ no unique solution to the problem for higher order approximation, e.g. quadratic equation

Conditional forecasts: Algorithm

- At time t , we
 - know past states $y_{1,t}, y_{2,t}$ and random realization of uncontrolled shocks $\varepsilon_{2,t}$
 - are given the desired values for controlled endogenous variables $y_{1,t}$
- We can solve the first block of the matrix equation to obtain:

$$\varepsilon_{1,t} = R_{11}^{-1} (y_{1,t} - T_{11}y_{1,t-1} - T_{12}y_{2,t-1} - R_{12}\varepsilon_{2,t}) \quad (4)$$

- Shock inversion requires matrix R_{11} to be square and full rank:
 - number of controlled shocks equal to number of constrained endogenous variables
 - controlled shocks must independently affect the controlled endogenous variables
→ no “redundant” shocks
- Given $\varepsilon_{1,t}$, unconstrained endogenous variables $y_{2,t}$ can be updated by evaluating the second block of equations:

$$y_{2,t} = T_{21}y_{1,t-1} + T_{22}y_{2,t-1} + R_{21}\varepsilon_{1,t} + R_{22}\varepsilon_{2,t} \quad (5)$$

Conditional forecasts: Dynare code I

NK_linear_controlled_forecast.mod

- Consider linear New Keynesian model with 3 exogenous shocks:

```
model(linear);  
  [name='Phillips Curve']  
  pie=beta*pie(+1)+kappa*y+eps_a;  
  [name='DIS curve']  
  y=y(+1)-1/sig*(inom-pie(+1))+eps_c;  
  [name='Nominal interest rate Taylor rule']  
  inom=phi_pi*pie+phi_y*y+eps_i;  
  [name='Law of motion for cost-push shocks']  
  eps_a=rho_a*eps_a(-1)+u_a;  
  [name='Law of motion for demand shocks']  
  eps_c=rho_c*eps_c(-1)+u_c;  
  [name='Law of motion for interest-rate shocks']  
  eps_i=rho_i*eps_i(-1)+u_i;  
end;
```

Conditional forecasts: Dynare code II

NK_linear_controlled_forecast.mod

- Set up stochastic shock distribution and compute decision rules at first order:

```
shocks;  
    var u_c  = 0.005^2;  
    var u_a  = 0.005^2;  
    var u_i  = 0.005^2;  
end;  
stoch_simul(order = 1, irf=0) y pie inom ;
```

Conditional forecasts: Dynare code III

NK_linear_controlled_forecast.mod

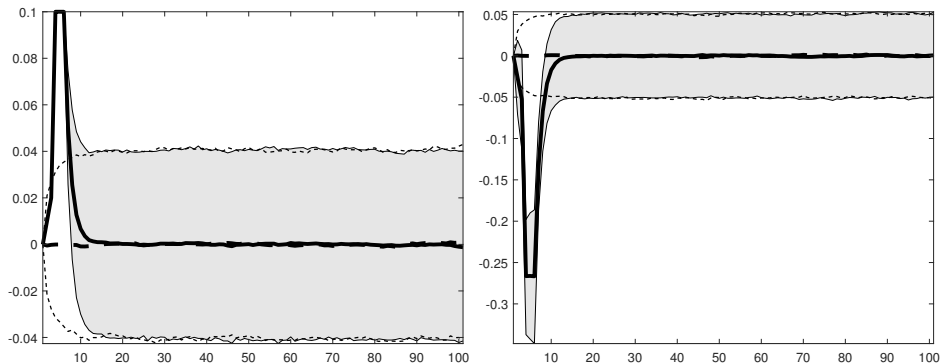
- Use `conditional_forecast_paths`; block to set nominal interest rate path over the next 5 periods
- Request forecast using `conditional_forecast` command, defining controlled shock set using the `controlled_varexo` option
- Plot the results using `plot_conditional_forecast` command

```
conditional_forecast_paths;
var inom;
periods 1 2 3:5;
values 0.01 0.02 0.1;
end;

conditional_forecast(parameter_set=calibration, controlled_varexo=(u_i));
plot_conditional_forecast(periods=10) inom y;
```

Conditional forecasts: Dynare output

NK_linear_controlled_forecast.mod



- Results are stored in the `oo_.conditional_forecast` field
- Plots show both conditional forecast (solid line) and unconditional forecast (dashed line) with `conf_sig%` confidence intervals

Dealing with unobserved states

- Forecasting scenarios usually involve conditioning on the state of the economy at the end of the available sample T
- Problem: while agents in RE models know the true state, econometrician does not
- Otherwise, we could simply put everything we need in the `histval` block as before
- Need to infer the relevant state variables x_T at the end of the sample from some data
→ filtering/smoothing problem
- Most commonly, this is done in a linear/first-order perturbation framework where well-known techniques can be efficiently exploited

Linear models: Kalman filter/smoothen

- For a given parameter set θ , the linear law of motion of states

$$x_{t+1} = Fx_t + w_{t+1} \quad (6)$$

implies:

$$x_{T+h|T} = F^h x_{T|T} + \sum_{j=1}^h F^{h-j} w_{T+j|T} \quad (7)$$

- In linear Gaussian world, the Kalman filter/smoothen naturally constructed optimal estimates of the states given observed data
→ yields initial condition $x_{T|T}$
- We could also directly use the Kalman filter recursions at the end of the sample to obtain the filtered estimates $x_{T+h|T}$ for $h > 0$

Calibrated smoother: Dynare code

`rbc_basic_calib_smoother.mod`

- Simulate artificial data and use it as the datafile to the `calib_smoother` command
- Request k -step ahead filtered variables and plot them

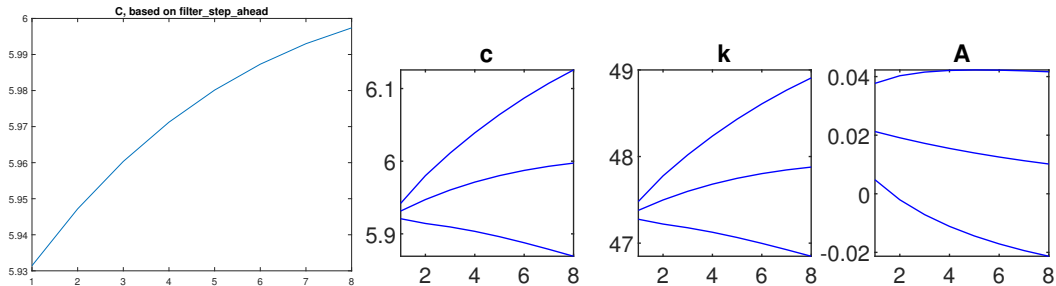
```
stoch_simul(order=1,irf=0,periods=200) c;

datatomfile ('rbc_data', {'c'});
varobs c;

calib_smoother(datafile='rbc_data',filter_step_ahead=[1:8]);
for var_iter=1:8
    forecast_c(var_iter)=oo_.FilteredVariablesKStepAhead
        (var_iter,varlist_indices('c',M_.endo_names),200+var_iter);
end
figure
plot(1:8,forecast_c);
title('C, based on filter_step_ahead','Interpreter','none')
```

Calibrated smoother forecasts: Code and output

rbc_basic_calib_smoother.mod



- Better way: use `smoother2histval` function to set `histval` block based on the smoothed states at time T from `calib_smoother`
- Use `forecast` command as before to compute conditional forecast:

```
smoother2histval;  
forecast(periods=8);
```


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Parameter uncertainty

- In many applications, parameter uncertainty is an important source of uncertainty for forecasts
→ comes on top of disturbance uncertainty
- We saw how to estimate the parameters of a DSGE model
→ characterized posterior distribution using MCMC
- Also saw how Monte Carlo integration over posterior allows evaluating integrals:

$$\mathbb{E} \left(g(\theta) | y^T \right) = \int g(\theta) p(\theta | y^T) d\theta \approx \frac{1}{S} \sum_{s=1}^S g(\theta_s) = \hat{g}_S \quad (8)$$

- As $g(\theta)$ can be any function of θ , it naturally encompasses forecasts $x_{T+h|T}$

Posterior forecasting: Dynare code

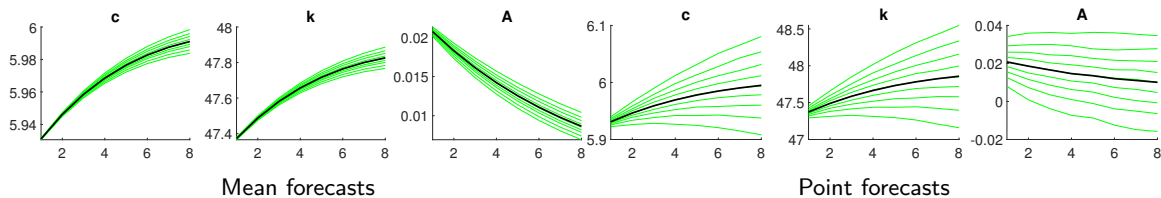
rbc_basic_posterior_forecast.mod

- Estimate the model parameters at first order
- Request 8-step ahead forecast

```
varobs c;  
  
estimated_params;  
rho, 0.9, beta_pdf, 0.7, 0.1;  
end;  
  
estimation(order=1, datafile=rbc_data, forecast=8,  
           mh_nblocks=1, mh_replic=4000, consider_all_endogenous);
```

Posterior forecasting: Dynare output

rbc_basic_posterior_forecast.mod



- So-called “mean” forecasts take only parameter uncertainty into account (zero shocks, i.e. shocks averaged out)
 - black line depicts mean across parameter draws
 - green lines depict the forecast deciles
- So-called “point” forecasts take both parameter and future disturbance uncertainty into account (randomly drawn shocks)

Recursive forecasting: Dynare code

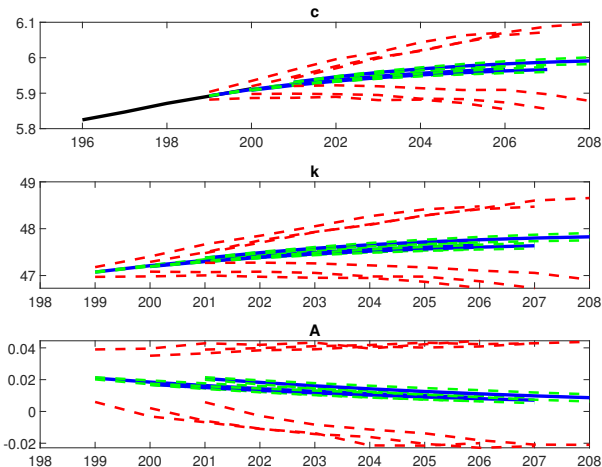
rbc_basic_recursive_estimation.mod

- Dynare also allows conducting recursive or **rolling window estimation and forecasting**
- Specifying a range for nobs in the estimation command will re-estimate the model for each specified observation and do forecasts, if requested

```
estimation(order=1,datafile=rbc_data,forecast=8,  
           mh_nblocks=1,mh_replic=4000,consider_all_endogenous,  
           nobs=[198:200]);
```

Recursive forecasting: Dynare output

rbc_basic_recursive_estimation.mod



- Blue solid lines: k -step ahead mean forecasts
- Black solid lines: actual data values for observables
- Green dashed lines: mh_conf_sig percent HPDI taking into account only parameter uncertainty
- Red dashed lines: mh_conf_sig percent HPDI with both parameter and future shock uncertainty

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Perfect foresight controlled paths

a.k.a. conditional forecast

- Occasionally, scenario analysis is less interested in quantifying uncertainty and more in preserving full nonlinearities
- Dynare 7 offers a new syntax for conditional forecasts in a perfect foresight context (supersedes deprecated `det_cond_forecast` command)
- `perfect_foresight_controlled_paths` block allows to flexibly define
 1. endogenous variables to be treated as exogenous
 2. exogenous variables to be treated as endogenousin a particular period
- Restriction: in each period, there need to be as many endogenized as exogenized variables (as many instruments as targets)
→ cannot (yet) have future shocks driving earlier controlled variable

Controlled paths: Dynare code I

rbc_basic_pf_controlled_paths.mod.mod

```
var c k A;
varexo epsilon;
...
perfect_foresight_controlled_paths;
    exogenize k;
    periods 2;
    values 50;
    endogenize epsilon;

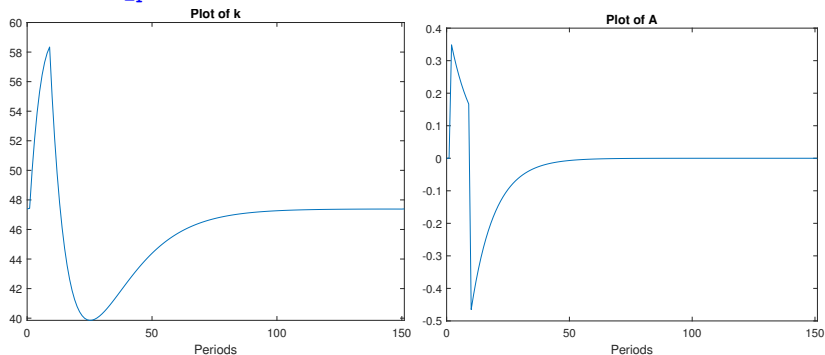
    exogenize k;
    periods 10;
    values 55;
    endogenize epsilon;
end;
```

Controlled paths: features and compatibility

- Can be combined with regular shocks or mshocks blocks
- Compatible with regular `perfect_foresight_solver` and `perfect_foresight_with_expectation_errors_solver`
- Not compatible with block decomposition and bytecode representation

Controlled paths: Dynare output

rbc_basic_pf_controlled_paths.mod



```
perfect_foresight_setup(periods=200);  
perfect_foresight_solver;  
rplot k;  
rplot A;
```

References I

- Bańbura, Marta et al. (2015). “Conditional forecasts and scenario analysis with vector autoregressions for large cross-sections”. *International Journal of Forecasting* 31 (3), 739–756.
- Beaudry, Paul and Franck Portier (2006). “Stock prices, news, and economic fluctuations”. *American Economic Review* 96 (4), 1293–1307.
- Smets, Frank and Rafael Wouters (2007). “Shocks and frictions in US business cycles: a Bayesian DSGE approach”. *American Economic Review* 97 (3), 586–606.